WEEKLY REPORT :

Project Title: Self Driving RC Car

Date: 10/20/2026

Instructor: Dr. Nguyen

TA: Ruben Ramirez

Project Members:

Ethan Wells (Group Leader)

Alexis Perez

Ethan Bishop

Abigail Duran

Weekly Summary (10/20/2026):

- This week the team tested the traffic-sign classifier with BuggyBot's own camera, studied ROS 2 for the vehicle software, and continued the custom PCB design. Ethan Wells completed last week's goal of printing the sign cards and photographing them with the Raspberry Pi Camera Module 3 on the CM5. He captured 10 of the 12 sign classes under three lighting conditions (no light, sunlight, and sunlight plus artificial light) and cut 329 sign crops with an OpenCV template-matching script. On those 329 camera crops the combined model classified every crop correctly (100% top-1), against 47.7% for the LISA model and 79.9% for the Mapillary model. The crops were cut with each sign's known class, so this measures sign classification and not automatic sign detection. He also reorganized the Hardware and Model Training folders, updated the maintained documentation, and repaired the links to the moved files. Alexis Perez continued his OpenCV learning and tested OpenCV on a webcam, ordered a car chassis so the vehicle components can be tested together, and wrote a TCRT5000 IR line-sensor script for the Raspberry Pi that reads the left and right course-line sensors on GPIO17 and GPIO27 and reports a lane action. Ethan Bishop installed ROS 2 under Windows Subsystem for Linux and drafted the node architecture for the vehicle: camera, sign detector, line sensor, IMU, behavior, motor driver, LCD, and teleop. He also drafted the data flow, in which

the behavior node turns lane, sign, and IMU data into driving decisions. Abigail Duran created an organized component checklist for the PCB design, reviewed the CM5 IO Board components and connections to decide which features are kept, removed, or modified, and added the IMU and LCD connectors to the board layout. Connecting the combined classifier to the CM5 camera and LCD pipeline, and measuring crops taken without using the known sign class, carry into next week.

Proposed Plan for Next Week:

- Ethan Wells: Connect the combined classifier to the CM5 camera and LCD pipeline, evaluate automatic sign crops that do not use the known sign class, and capture new setups at measured distances and angles while recording accuracy and inference time.
- Alexis Perez: Implement OpenCV with the Raspberry Pi camera.
- Ethan Bishop: Create a test program for the ROS 2 implementation.
- Abigail Duran: Continue developing the PCB design and component placement, and work toward finalizing the PCB design for manufacturing.

Weekly Contributions:

- Ethan Wells:
    - Photographed the printed sign cards with the CM5 camera: 10 sign classes under three lighting conditions, cut into 329 sign crops with an OpenCV template-matching script (5 hours)
    - Scored the LISA, Mapillary, and combined models on the same 329 camera crops (combined 329 of 329 correct, LISA 47.7%, Mapillary 79.9%)
    - Reorganized the Hardware and Model Training folders, updated the maintained documentation, and repaired links to the moved files (1 hour)
    - Prepared the weekly slides and report (1 hour)
- Alexis Perez:
    - Continued the OpenCV course and tested OpenCV on a webcam (2 hours)
    - Ordered a car chassis to test the vehicle components together (1 hour)
    - Wrote a TCRT5000 IR line-sensor script for the Raspberry Pi that reads the left and right course-line sensors and reports a lane action (3 hours)
- Ethan Bishop:

    - Installed ROS 2 under Windows Subsystem for Linux and drafted the ROS 2 node structure for the camera, sign detector, line sensor, IMU, behavior, motor driver, LCD, and teleop (4 hours)
    - Drafted the ROS 2 data flow in which the behavior node turns lane, sign, and IMU data into driving decisions (2 hours)
- Abigail Duran:
    - Created an organized component checklist/spreadsheet for the PCB design (2 hours)
    - Reviewed the CM5 IO Board components and connections to determine which features should be kept, removed, or modified (2 hours)
    - Added the IMU and LCD connectors to the board layout (2 hours)

Hour Tracker:

Name	Hours For the Week	Cumulative Hours
Ethan Wells	7	48
Alexis Perez	6	41
Ethan Bishop	6	40
Abigail Duran	6	41